

Deep learning models on wearable biosensor data for predicting clinical deterioration and reducing hospital admissions in chronic heart failure patients.

Healthcare × Remote patient monitoring × Machine learning · OS011 v2 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
52 is the world moving	64 can we play, can we win	MEDIUM 0 named in the evidence	€174m partial evidence · per year	LATER research and standards activi...	L1 14 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is a remote patient monitoring service for chronic heart failure patients, using AI on wearable biosensor data to predict clinical deterioration and reduce hospital admissions. It is for healthcare providers and payers facing high readmission rates and emergency department strain. The need is urgent because evidence shows RPM can detect early decompensation, yet current solutions are fragmented and lack robust clinical validation.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€282m €23.1m – €1.3bn	€174m €14.2m – €827m	€4.9m €398k – €23.1m	partial
Observed: tendered contract value, annualised	€70.8m	€70.8m	€2.0m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Healthcare (EU27_2020)	46,762 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Machine learning	5.1 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_ain2 · E_AI_TML	2025
Annual value of one engagement	€970k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-07-22
Size-weighted buyer base	5,698 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	2.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Enterprises using machine learning for data analysis Part of this vertical is not covered separately by the enterprise ICT survey, so the all-activities rate was used for those slices. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 2 of 2 declared geographies are outside the European reference data (MY, OM); the estimate covers the European footprint only. Sector adoption is read from Eurostat's all-activities aggregate for part of this vertical, which the enterprise ICT survey does not cover separately: PROXY — ICT survey excludes health activities; PROXY — residential care

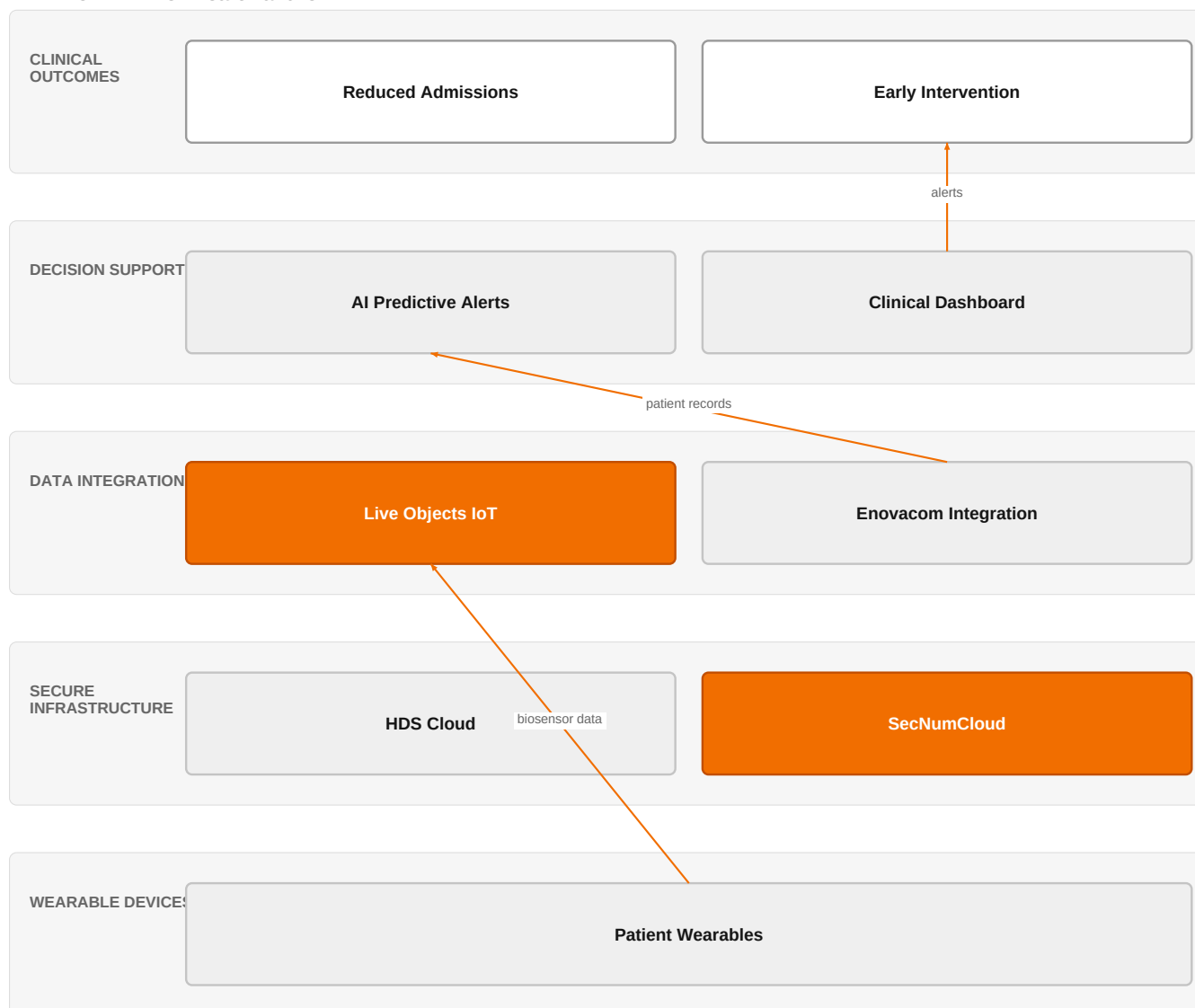
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€70.8m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-07-22
Assumed obtainable share of the serviceable market	2.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 7 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 2 of 2 declared geographies are outside the European reference data (MY, OM); the estimate covers the European footprint only.

The solution, in one picture

AI-Driven RPM for Heart Failure



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how wearable data flows through Orange's secure IoT and AI platform to deliver actionable clinical alerts that reduce hospital admissions.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Regulatory and clinical acceptance of remote patient monitoring is growing, with CE-certified wearables and AI-enabled monitoring being studied in trials like SMART-CARE. Technology has matured: IoT and AI frameworks now enable real-time monitoring for elderly and chronically ill patients, and wearable devices can continuously measure physiological signals. However, clinical adoption is still limited by concerns about interpretability of deep learning models and the need for robust evidence, as highlighted by systematic reviews and meta-analyses.

Sources: SIG-E95AED1CBD71 openalex.org 2025-12-10; SIG-0205DE7D82FE openalex.org 2026-04-01; SIG-414D1342F56A openalex.org 2026-03-25; SIG-373E00524578 openalex.org 2025-08-29; SIG-DBBEFE8CCC78 openRxiv 2026-02-27

Who buys, and why now

The buyer is typically a chief medical officer or head of cardiology at a hospital or health system, while the pain is felt by care teams managing heart failure patients who are frequently readmitted. The trigger is often a high 30-day readmission

rate or emergency department overcrowding, prompting investment in remote monitoring to reduce hospitalizations and improve outcomes.

Why it is hot now — every claim bound to a dated source

- An IoT and AI integrated framework allows real-time health monitoring for elderly and chronically ill patients.
[SIG-0205DE7D82FE]
- AI enhances signal processing and predictive insights in wearable biosensors for personalized healthcare.
[SIG-548EBD3DF0D9]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its IoT platform Live Objects to securely collect data from wearable biosensors, and Enovacom/Enovalife to integrate with hospital information systems. AI/Data experts would build predictive models on the data, leveraging Orange's healthcare expertise and research capabilities. The solution would be hosted on Orange's certified infrastructure (HDS, SecNumCloud, ISO 27001) to ensure compliance. Partners like NVIDIA could provide GPU acceleration for model training, and AWS or Google Cloud could offer scalable compute, but Orange would orchestrate the end-to-end service.

Why Orange can win

Orange can win because it combines healthcare-specific certifications (HDS, SecNumCloud) with a full-stack capability: IoT connectivity via Live Objects, healthcare integration via Enovacom, and AI expertise from its data and research teams. This is a rare combination among telecom operators, and Orange's sovereign cloud credentials align with healthcare data privacy requirements. However, the assets are not unique; hyperscalers also offer strong AI and cloud, so Orange must differentiate on trust and integration.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer provides the technology in a matching domain	unconfirmed
Enovacom / Enovalife	offer	L1	offer addresses the use case but not this technology	unconfirmed
AWS	partner	L2	partner provides the required technology	unconfirmed
Google Cloud	partner	L2	partner provides the required technology	unconfirmed
NiCE	partner	L2	partner provides the required technology	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
HDS (Hébergeur de Données de Santé)	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Healthcare experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
MicroPort CardioFlow	reference	SUP	published reference in this vertical for this use case	unconfirmed
Bliksund	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 7.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Machine learning; competes in OX: Smart Industries — also an Orange partner	structural	—
Google Cloud	Hyperscaler	sells Machine learning — also an Orange partner	structural	—
Atos / Eviden	Global systems integrator	sells Machine learning; active in Healthcare	structural	—
NICE	Customer-experience platform	sells Machine learning — also an Orange partner	structural	—
Databricks	Data and AI platform	sells Machine learning	structural	—
Dataiku	Data and AI platform	sells Machine learning	structural	—
Palantir	Data and AI platform	sells Machine learning; active in Healthcare; competes in OX: Smart Industries	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	active in Healthcare; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is your current 30-day readmission rate for heart failure patients, and how does it compare to national benchmarks?
- 2 Do you already use any remote monitoring devices or platforms for heart failure patients? If so, what are the main gaps?
- 3 How do your clinicians currently detect early signs of decompensation? Is it reactive or proactive?
- 4 What is your biggest barrier to adopting AI-based monitoring: clinical trust, data integration, or regulatory compliance?
- 5 Are you prepared to invest in a multi-year program to build and validate predictive models, or do you need a turnkey solution?

Objections you will meet

They will say	You can answer
We already have a remote monitoring program with basic alerts; why do we need AI?	That's true, many programs use simple thresholds. But AI can detect subtle patterns that thresholds miss, potentially reducing false alarms and catching deterioration earlier. Our approach builds on your existing infrastructure.
We're concerned about the accuracy and interpretability of AI models.	Valid concern. We focus on explainable AI, and we will work with your clinicians to validate models on your data. We also start with pilot studies to build trust.
How do you ensure data privacy and compliance?	We host on HDS-certified infrastructure and comply with ISO 27001 and SecNumCloud. We also sign data processing agreements and ensure end-to-end encryption.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a remote patient monitoring tender that highlights our HDS and SecNumCloud certifications, Live Objects and Enovacom/Enovalife offers, and partnerships with AWS and NVIDIA to support AI-driven predictive analytics on wearable biosensor data.
Sales	In a conversation with a healthcare provider's digital health lead, say: 'We can leverage our Live Objects IoT platform and Enovacom/Enovalife healthcare solutions, backed by our HDS and SecNumCloud certifications, to explore remote monitoring for chronic heart failure patients.'
Strategist / Innovator	Commission a deep-dive brief on the feasibility of combining Live Objects and Enovacom/Enovalife with AI/Data/Cloud and IoT experts to predict clinical deterioration from wearable biosensor data, focusing on HDS and SecNumCloud compliance for healthcare data.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-AFC3B44CAE18	2026-05-22	Global Cardiology Science...	1	Remote Patient Monitoring in Heart Failure
SIG-0205DE7D82FE	2026-04-01	openalex.org	1	ZigBee Based Low Latency IoT and AI Integrated Framework for Real Time Telehealth Monitoring
SIG-414D1342F56A	2026-03-25	openalex.org	1	Wearable devices and cardiovascular health: revolutionizing remote monitoring and disease preve...
SIG-548EBD3DF0D9	2026-03-20	openalex.org	1	Smart wearable and implantable biosensors for continuous health monitoring: materials, biocompa...
SIG-A3F10F84C60F	2026-03-01	APL Electronic Devices	1	Industry perspective: Wearable electronic devices for remote patient monitoring
SIG-DBBEFE8CCC78	2026-02-27	openRxiv	1	Remote Patient Monitoring in Heart Failure: A Systematic Review, Meta-Analysis, and Trial Seque...
SIG-E95AED1CBD71	2025-12-10	openalex.org	1	Artificial intelligence-based remote monitoring for chronic heart failure: design and rationale...
SIG-7255230B0C13	2025-11-28	2025 3rd DMIHER Internati...	1	Integrating IoT, Machine Learning, and Federated Learning for Early Cardiovascular Disease Dete...
SIG-4016692CAFB1	2025-10-21	Cureus	1	Reducing Emergency Department Strain Through Usable Wearables: A Proof-of-Concept for Multi-sen...
SIG-5A5443D644A0	2025-10-18	openalex.org	1	Invasive and Non-Invasive Remote Patient Monitoring Devices for Heart Failure: A Comparative Re...
SIG-605D6DC17233	2025-09-12	openalex.org	1	AI-Driven Real-Time Monitoring of Cardiovascular Conditions With Wearable Devices: Scoping Revi...
SIG-373E00524578	2025-08-29	openalex.org	1	Personalized health monitoring using explainable AI: bridging trust in predictive healthcare
SIG-62BCB30A5034	2025-08-29	openalex.org	1	A Comprehensive Survey on Wearable Computing for Mental and Physical Health Monitoring
SIG-5EC51EEB6EC8	2025-08-21	openalex.org	1	From Clinic to Cloud: Efficacy of AIAssisted Remote Monitoring of Patients With Implantable Ca...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); risks_and_unknowns (missing from the model output); diagram (4 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:40:57+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.